To implement this functionality, we'll introduce two new helper functions: `streamUI` and `createStreamableUI`. These functions provide the core functionality for vibrant UI generation based on AI responses.

4.1.5 Creating streamable UI components from LLM providers with streamUI

The `streamUI` function is an effective tool that allows you to generate and stream UI components based on responses from LLM providers. It orchestrates the process of generating UI components, allowing for active and incremental UI updates. The function takes user input, sends it to the LLM, and based on the response, can either render text directly as a component or call specific tools to generate more complex UI components. The resulting components are then streamed to the client in real time.

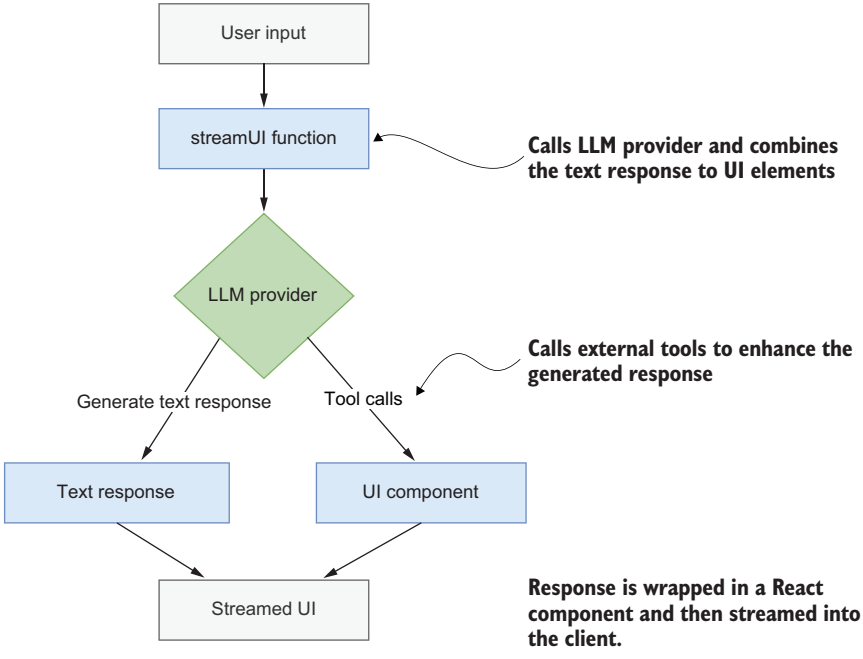


Figure 4.6 Example of how `streamUI` works. This function orchestrates the process of generating UI components based on LLM responses. It handles both simple text responses and more complex tool calls and then streams the fully rendered components as React nodes into the client.

The `streamUI` function takes user input and sends it to the LLM provider. Based on the LLM's response, it can either render text directly as a component or call specific tools to generate more complex UI components. This allows components to be streamed to the client in real time.

To understand how it works in practice, see the following listing, where we render a single generated prompt response directly as a `<Chat />` message component.